6502 Monitor – ESP32 Pin Mapping (NodeMCU DevKit V1)

Atari ESP32 Project

June 12, 2026

1 Data Bus (Bidirectional)

6502 Signal	**ESP32 GPIO**	**Board Label**
D0–D7	GPIO 4, 5, 13, 14, 16, 17, 18, 19	D4, D5, D13, D14, RX2, TX2, D18, D19

2 Address Bus (A0–A7 decoded; A8–A10 not wired)

6502 Signal	**ESP32 GPIO**	**Board Label**	
A0–A3	GPIO 34, 35, 36, 39	D34, D35, VP, VN	(**Input Only**)
A4–A7	GPIO 32, 33, 21, 27	D32, D33, D21, D27	

3 Control Signals

Signal	**GPIO**	**Label**	**Dir**	**Description**
PHI2	GPIO 2	D2	IN	6502 System Clock (1.79 MHz)
R/W	GPIO 15	D15	IN	Read/Write
SEL_N	GPIO 22	D22	IN	$D1XX / CCTL selection (Active Low)
ROMSEL	GPIO 23	D23	IN	$D800–$DFFF range (Active Low)
EXTSEL	GPIO 3	**RX0**	OUT	Disable Atari internal memory (Active Low)
MPD	GPIO 0	**BOOT**	OUT	Math Pack Disable (Active Low)

4 Notes

- **A8–A10 not wired:** the 256-byte ROM image mirrors 8× across the 2 KB $D800–$DFFF range.
- **GPIO 0 (MPD / BOOT):** kept HIGH during boot by the Atari pull-up (isolated via TXS0108E).
- **GPIO 3 (EXTSEL / RX0):** shared with UART RX; not usable for serial input at runtime.
- All signals pass through **TXS0108E** bidirectional level shifters (3.3 V ↔ 5 V).